

MAT 115E Introduction to Programming Language

Lab-4/ CRN : 23492

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1 Question 1

Write a C source code that accomplishes the followings:

A USER DEFINED FUNCTION:

- Write a **C function** named **isAlphabet** that takes a character as a parameter and tests whether or not the character is alphabet or not. A character is considered to be an alphabet if it is between **a-z** or **A-Z**. Your function returns 1 if the character is alphabet and 0 if not.

MAIN FUNCTION:

- Generate a random character between the values 32 and 126 in ASCII table. (only printable characters)
- Pass this random character to the function you coded.
- Print the returned value from the user defined function.

Sample Output:

We assume that '*' is the random character generated by program. User defined function is calling for the character '*' in main function. Then, result will be displayed in main function in the following form:

> * is alphabet or not : 0

2 Question 2

Write a **C function** named **hasMidpoint** that accepts three integers as parameters and returns 1 if one of the integers is the midpoint between the other two integers; that is, if one integer is exactly halfway between them. Your function should return 0 if no such midpoint relationship exists.

Calls such as the following should return 1: hasMidpoint(4, 6, 8); hasMidpoint(8, 8, 8); hasMidpoint(25, -5, 10).

Calls such as the following should return 0: hasMidpoint(1, 2, 1); hasMidpoint(21, 9, 58); hasMidpoint(1, 2, 1).

3 Question 3

Write a C source code that accomplishes the followings:

A USER DEFINED FUNCTION:

- Write a **C function** that takes three edge lengths of the triangle as parameters and tests whether or not these three edge lengths form a triangle.

MAIN FUNCTION:

- Generate three numbers between 3 and 10 using C built-in function rand().
- Pass those numbers to the user defined function.
- Print the returned value from the user defined function.

Sample Output:

We assume that three edge lengths generated randomly by program are 6, 3, and 8, respectively. Your function should return 1 if you pass those values to the function. In main function, you can print the simple message as below:

> Do these wires form a triangle ? → 1